

Math4Wisdom

July 1, 2026

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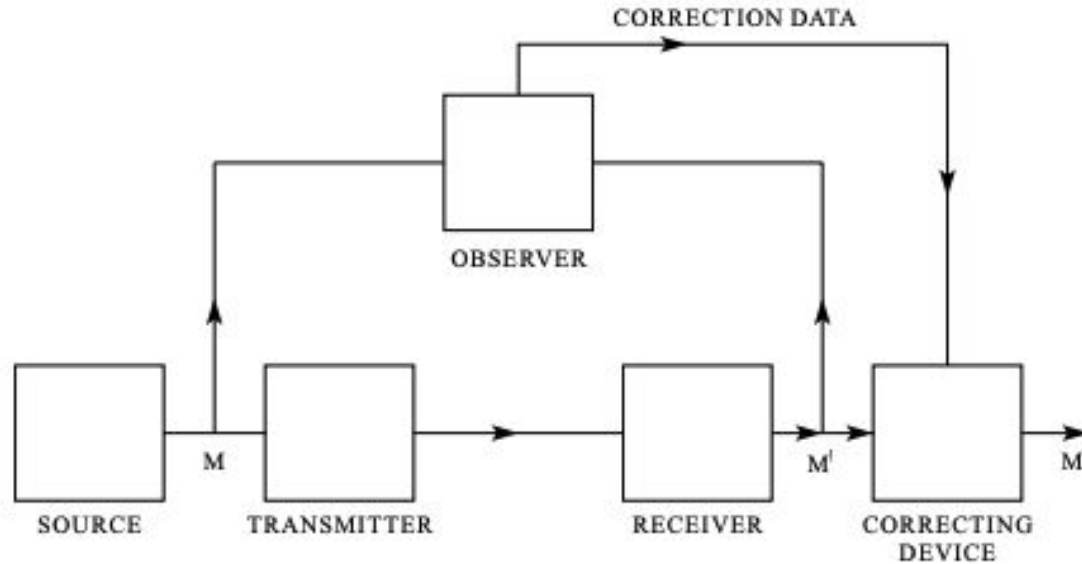


WOLFRAM
BLOCKCHAIN LABS

Questions

- What is the logical process of change which occurs in physical systems?
- What is the same everywhere regardless of physical systems?
- What language is the same everywhere?
- What does $2 + 2 =$
- What does $73 - 42 =$
- What does $5 \times 26 =$
- What does $101 / 43 =$
 - 2.3488372093
- What does a \$\$ cost to make?

Mathematical Theory of Communication - C.Shannon



Theorem 17: The capacity of a channel of band W perturbed by white thermal noise power N when the average transmitter power is limited to P is given by

$$C = W \log \frac{P+N}{N}.$$

What is seigniorage?

The value of
money

—

The cost of
producing
money

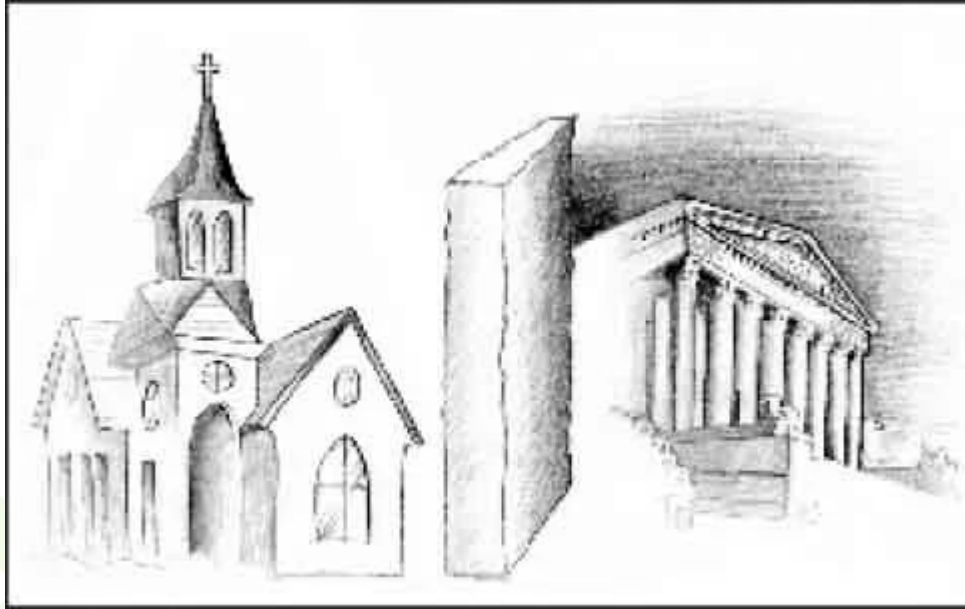
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Seigniorage

Seigniorage actualized: \$0.959 - \$0.9999

Denomination	Variable Printing Costs
\$1 and \$2	4.1 cents per note
\$5	7.1 cents per note
\$10	6.8 cents per note
\$20	7.3 cents per note
\$50	N/A ¹
\$100	11.3 cents per note

Separate Money from State

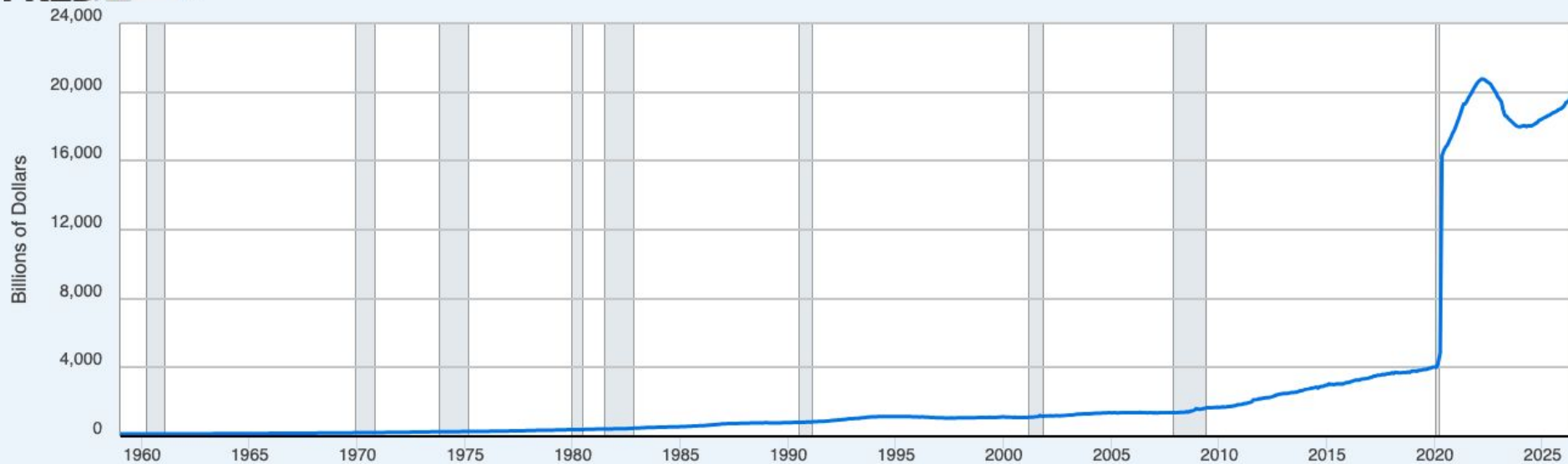


- Must tap fundamental ideas:
 - Physics
 - Information Theory
 - Distributed Systems
 - Economics
- Metacrisis
 - Generative functions
 - Arms races
 - Tragedy of Commons
 - Chaos or Tyranny?

“Must erect a wall separating church and the state for the benefit of free society.”

Monetary Supply M1

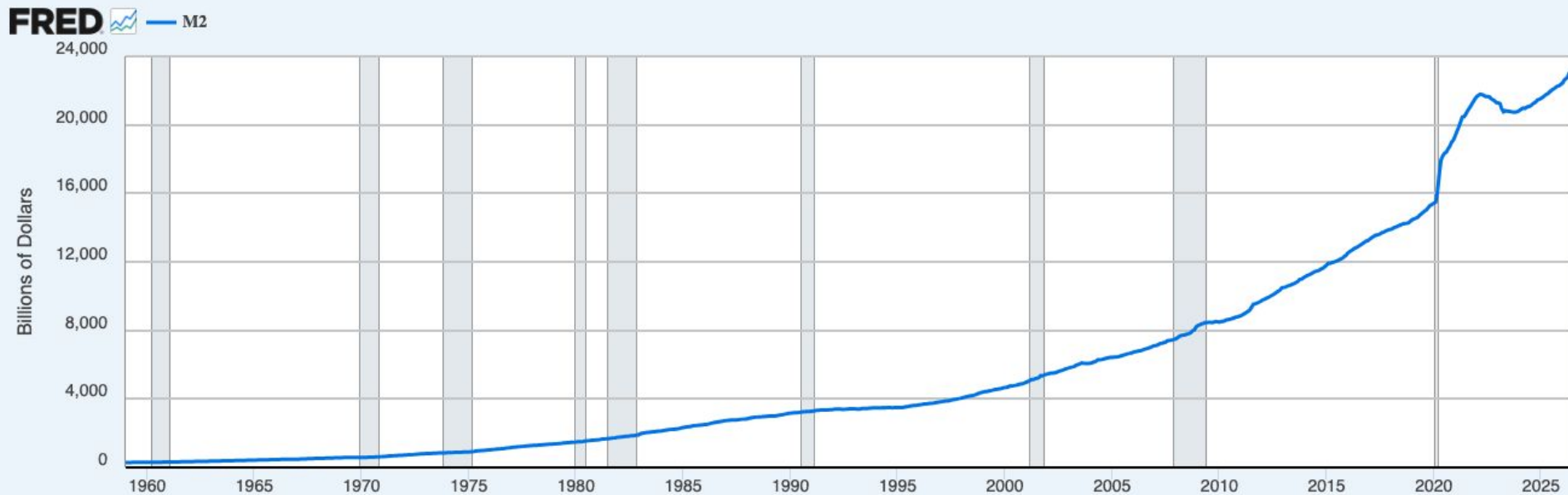
FRED  — M1



Source: Board of Governors of the Federal Reserve System (US) via FRED®
Shaded areas indicate U.S. recessions.

fred.stlouisfed.org

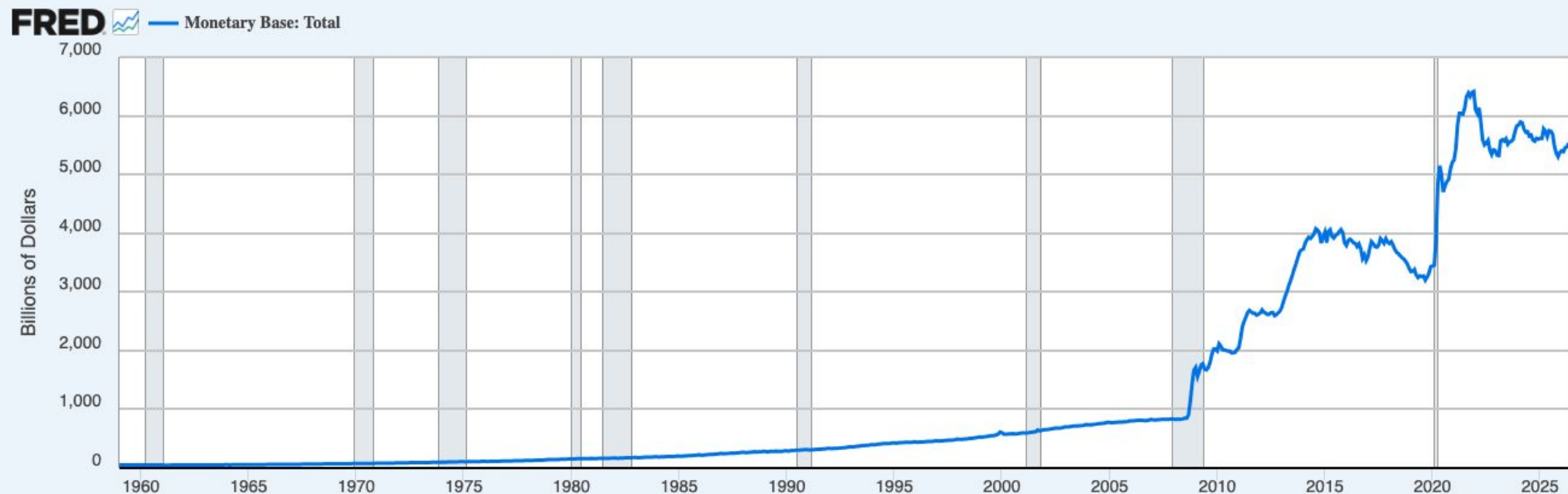
Monetary Supply M2



Source: Board of Governors of the Federal Reserve System (US) via FRED®
Shaded areas indicate U.S. recessions.

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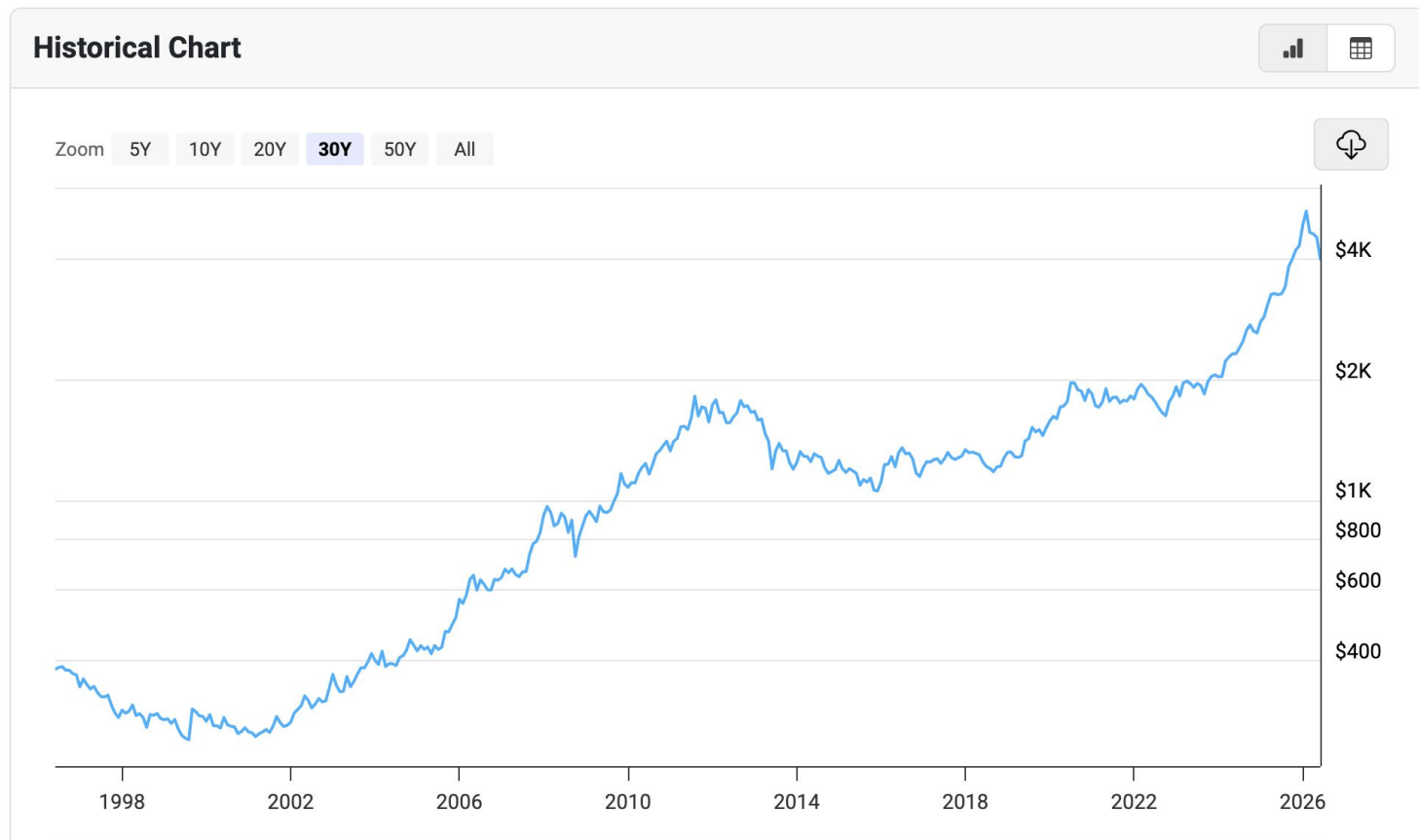
Total Monetary Supply



Source: Board of Governors of the Federal Reserve System (US) via FRED®
Shaded areas indicate U.S. recessions.

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Up 2,000% since 25 years ago



Bitcoin Production Function

The diagram illustrates the Bitcoin Production Function equation, which calculates the total number of bitcoins produced over time. The equation is:

$$\sum_{i=0}^{32} 210,000 * \left\lfloor \frac{50}{2^i} \right\rfloor$$

The components of the equation are annotated as follows:

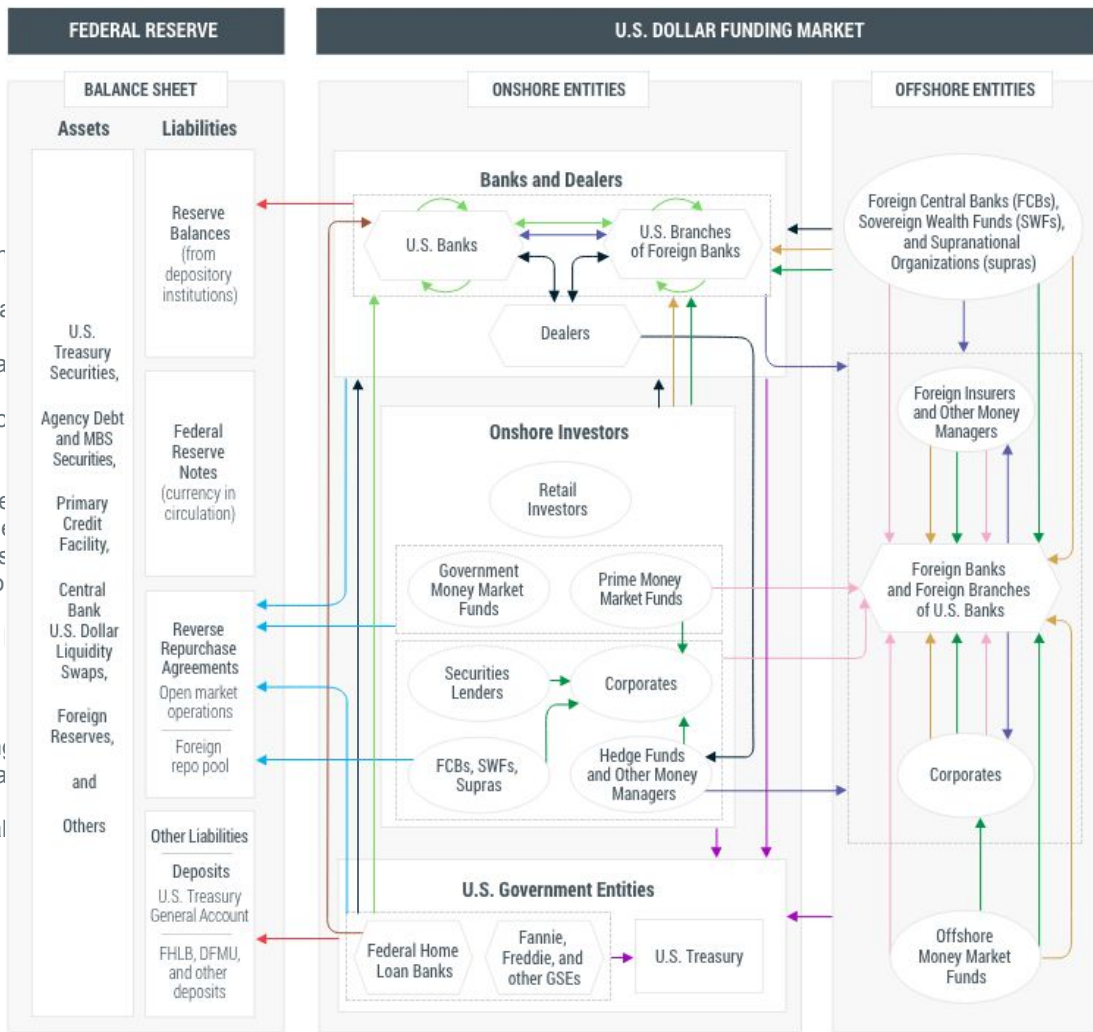
- 32**: The total number of halving epochs.
- $\sum$: The symbol for a summation equation.
- $i=0$: Initial Epoch.
- 210,000**: The # of blocks in each halving epoch.
- 50**: The maximum (initial) block reward.
- 2^i : The Halving.
- $\left\lfloor \frac{50}{2^i} \right\rfloor$: The current halving epoch.

Payment Gateway Transaction Process

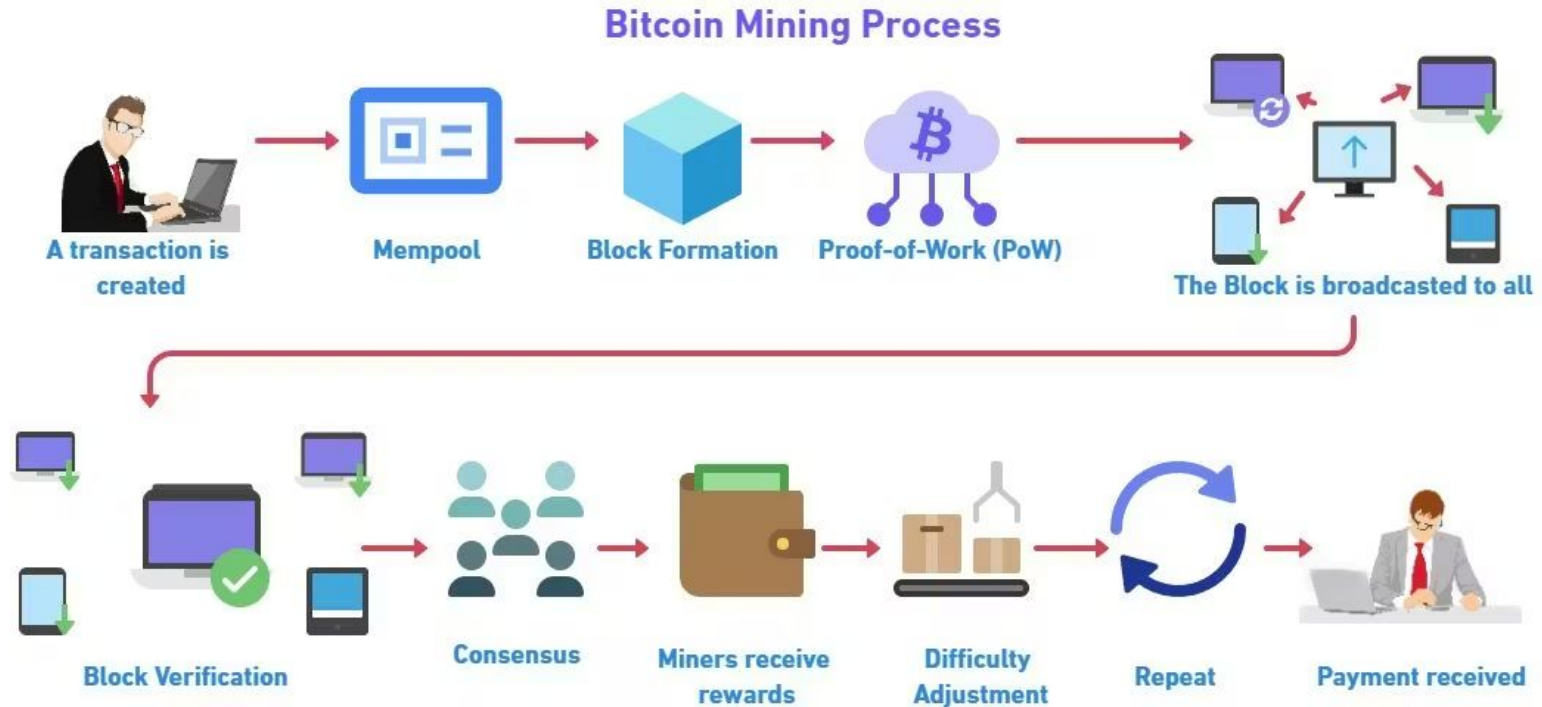


NY Fed Map

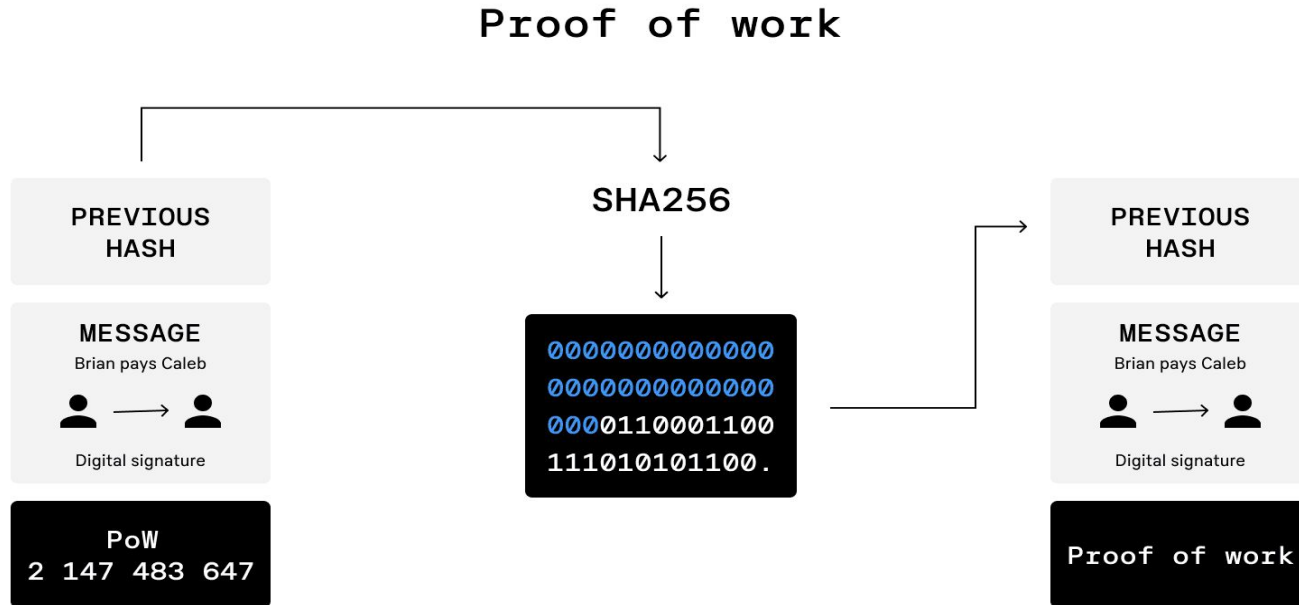
- **Agency MBS:** Mortgage-backed securities issued by government enterprises such as Fannie Mae or Freddie Mac
- **DFMU:** A designated financial market utility, such as the Clear Payments Company or the Chicago Mercantile Exchange
- **Eurodollars:** U.S. dollar-denominated deposits at foreign banks outside of the United States
- **Fannie:** The Federal National Mortgage Association, commonly known as Fannie Mae
- **FCB:** Foreign central bank
- **Foreign repo pool:** Overnight U.S. dollar investment service the Federal Reserve offers to foreign official and international accounts, and they use to swap overnight cash balances for Federal Reserve securities
- **Freddie:** The Federal Home Loan Mortgage Corporation, commonly known as Freddie Mac
- **GSEs:** Government-sponsored enterprises, such as Fannie Mae and the Federal Home Loan Banks
- **Intermediary institutions:** Institutions that both borrow and lend in U.S. dollar funding markets
- **Investor institutions:** Institutions that invest in dollar funding markets
- **MBS:** Mortgage-backed securities guaranteed by Fannie Mae or Freddie Mac
- **Supra:** Supranational organization, such as the International Monetary Fund or the United Nations
- **SWF:** Sovereign wealth fund



Bitcoin's Proof of Work

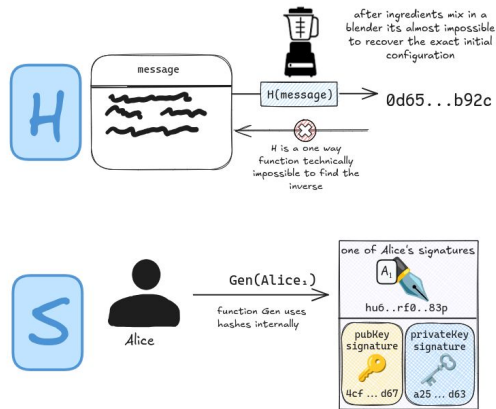


Proof of Work

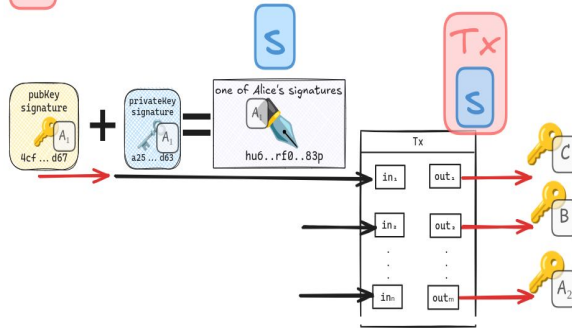


Core Ingredients of Blockchains

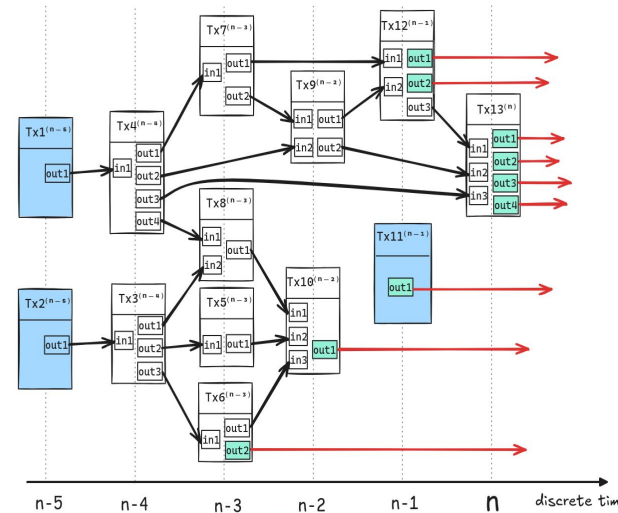
① H S



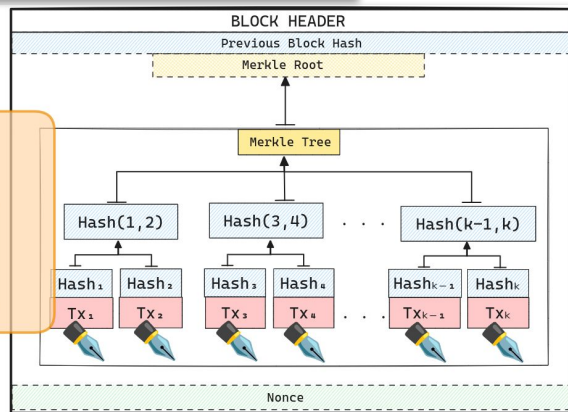
② Tx



③ UTXO set
Ledger

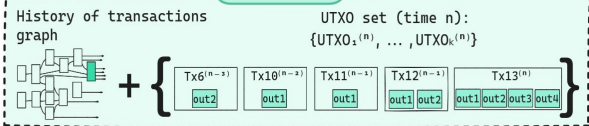
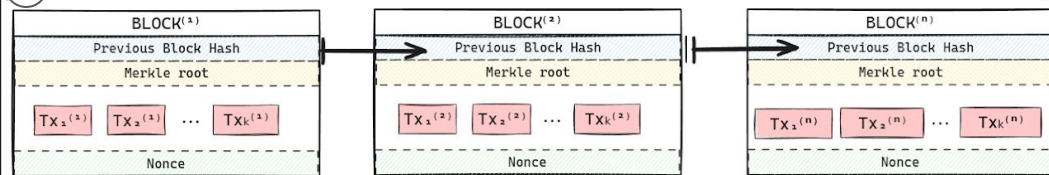


④

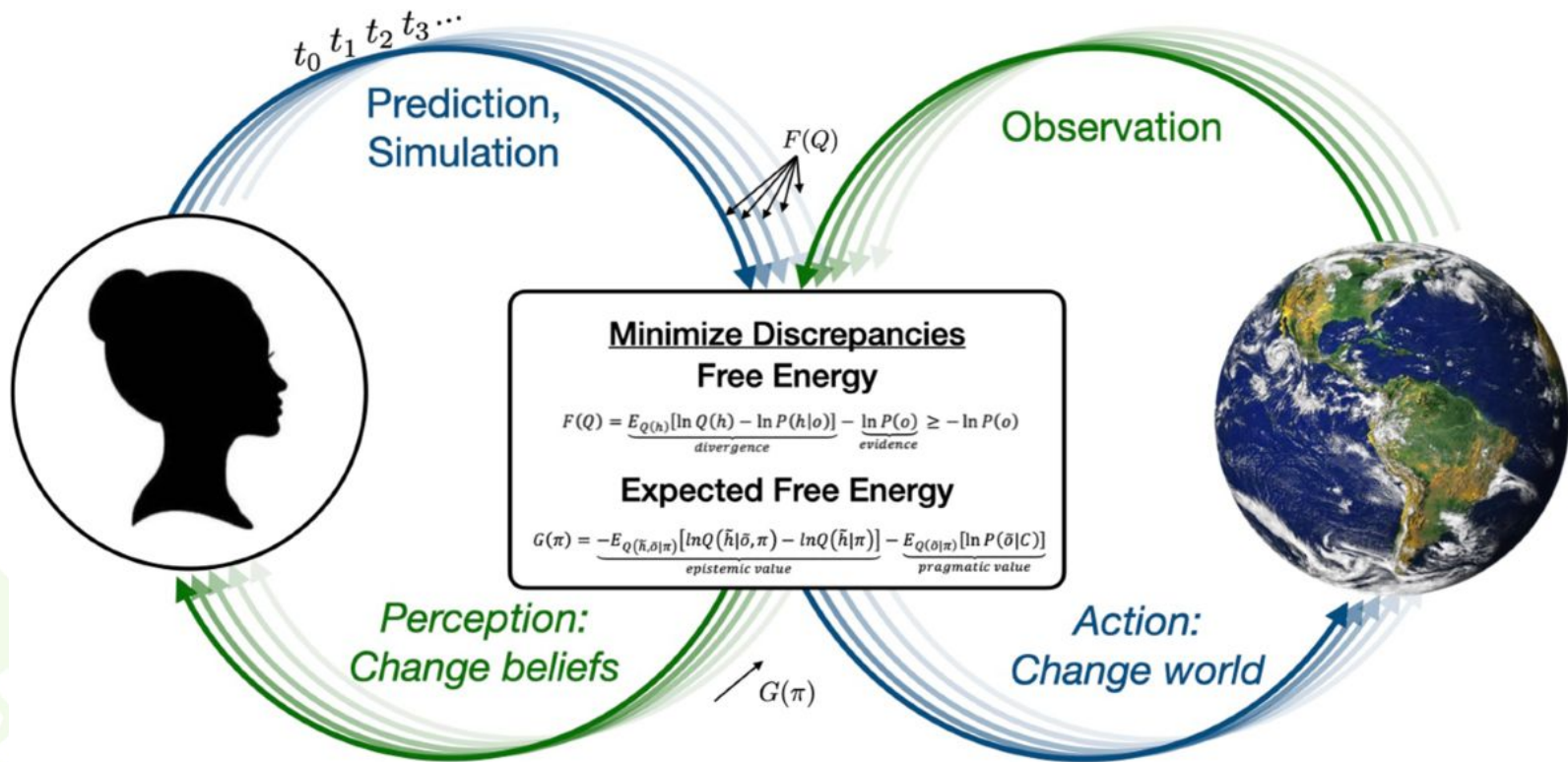


1. Hashes
2. Signatures
3. TX Ledger
4. Blocks
5. Chain

⑤



Active Inference



Normalization + Standardization

1. Source: Oil $\neq$ Solar $\neq$ Nuclear
2. Non-Fungibility: A watt here $\neq$ A watt there
3. Commodification of Compute: FLOP here $\approx$ FLOP there
4. Algorithmic Commodification: What is provably equivalent?
5. Characteristics of Randomness: Permissionless, Enforceable, Democratic, Meritocratic, Private, Definable (Precise, Open-Source, etc)

Simple Rule, Emergent Results

Proof of Entropy Minima

$$\triangle S = 1/2^n$$

of states removed

SHA256



```
0000000000000000
0000000000000000
0000110001100
111010101100.
```

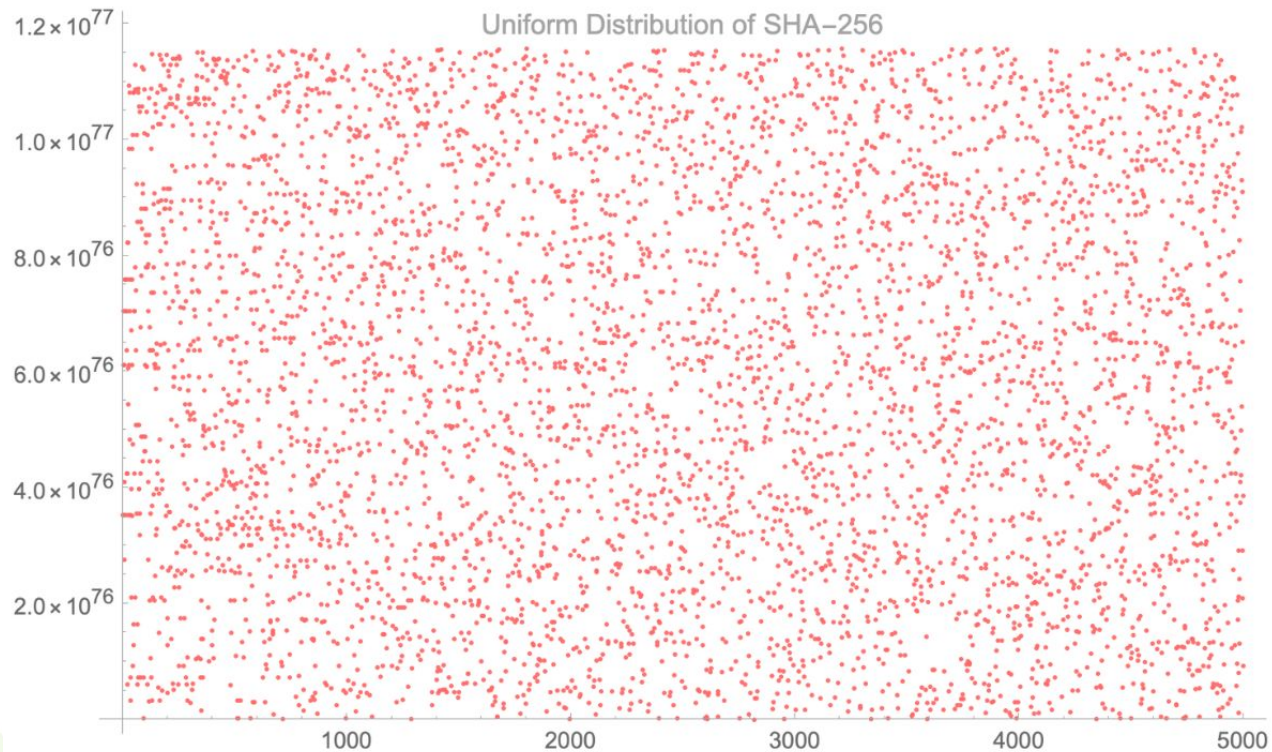
"A difference that makes a difference."

The highlighted red zero is a crucial, deterministic factor in the measurement of work (via statistical entropy) for blockchain consensus mechanisms. This single bit, which is forfeited in traditional PoW (longest chain rules) is employed in PoEM (heaviest chain rules), and is the difference between achieving instantaneous finalization and waiting for multiple rounds of validation. It is the difference between excess thermodynamic waste, and precise measurement of free energy expenditure. It is intrinsic to the system itself, which resolves two valid competitive outputs without conflict. This means that consensus can be established without contention, reaching finality as fast as physically possible. This forms the basis of sharded (scalable) proof of work system and the characteristic known as "causal invariance."

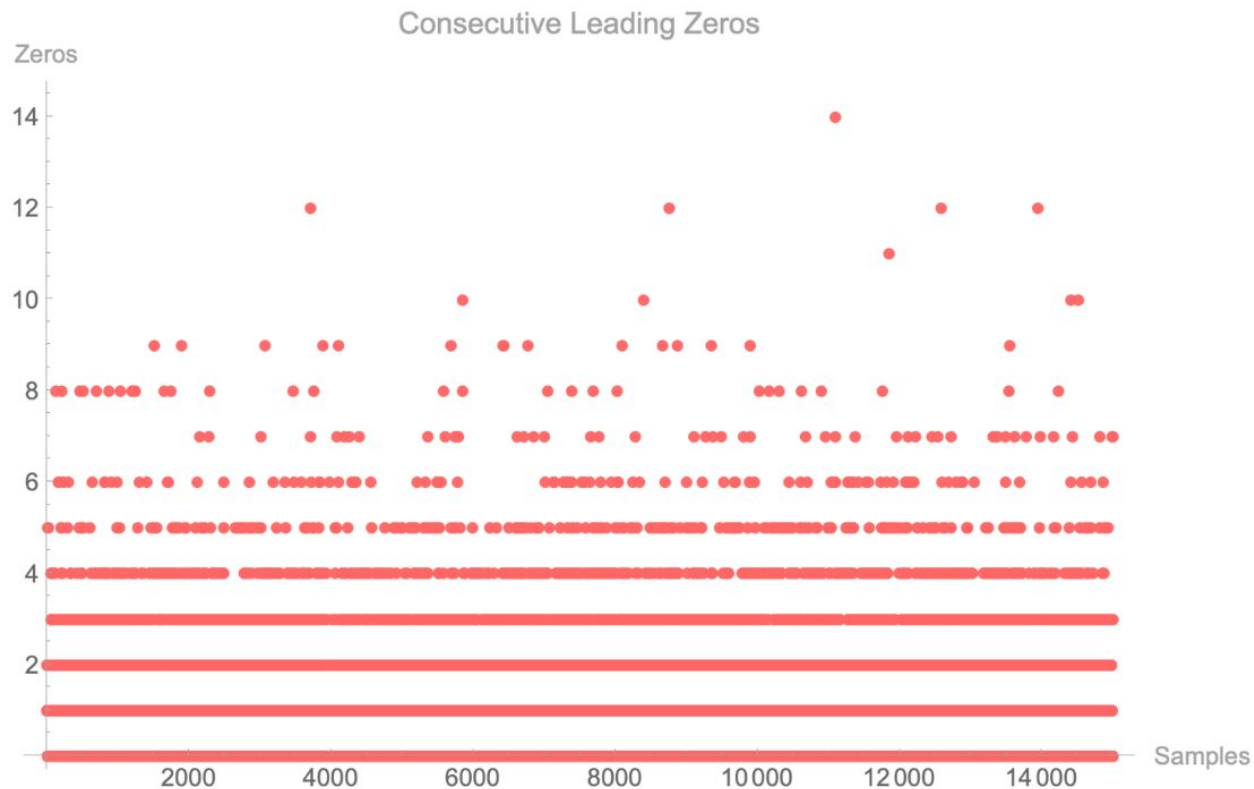
Proof of (Shannon) Entropy Minimization

- The total possible state space of a SHA-256 output = 2^{256}
- Rule: # of consecutive leading 0s in SHA-256 output (beyond arbitrary threshold)
- Example:
 - 0000000010110100111010101110011011011010110000100000010010111111111110000011
000010101001010110101001010110011001000111001001111010000010010110101010011001
011001010111001011011010100110001010001011100101010001110011110
- Interpretation: # of bits of uncertainty removed (accumulated/minimized)

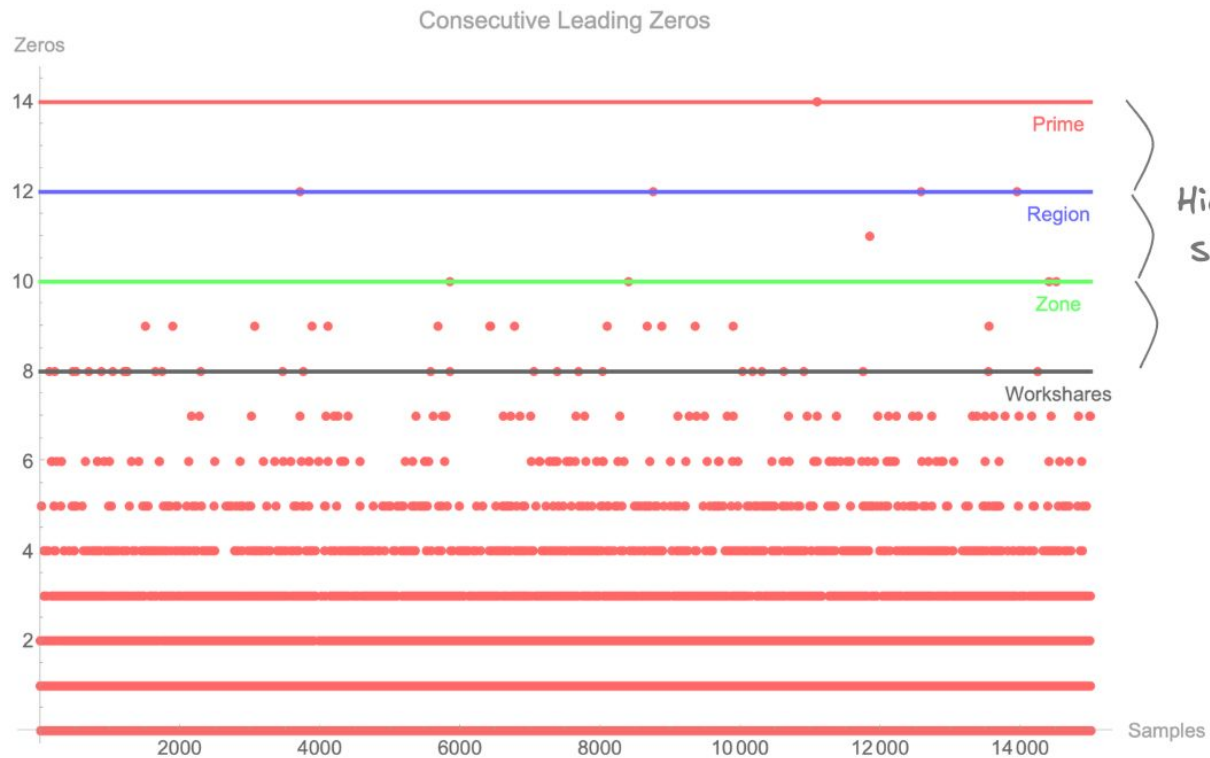
Maximum Entropy



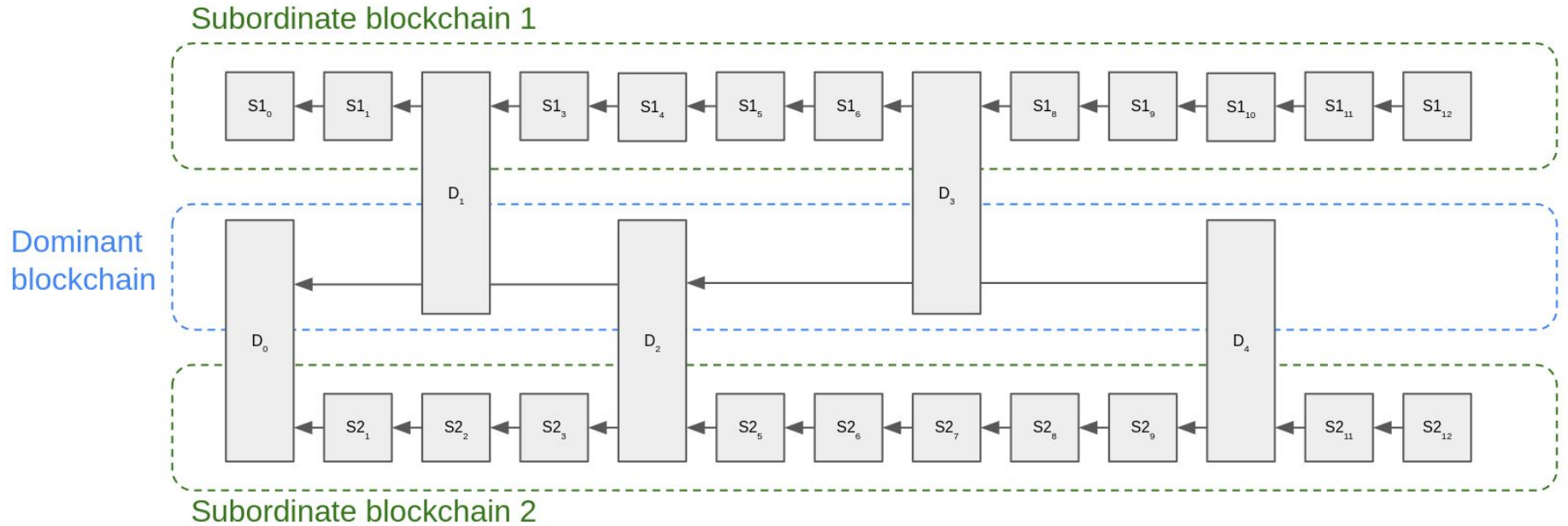
Maximum Entropy + Constraint



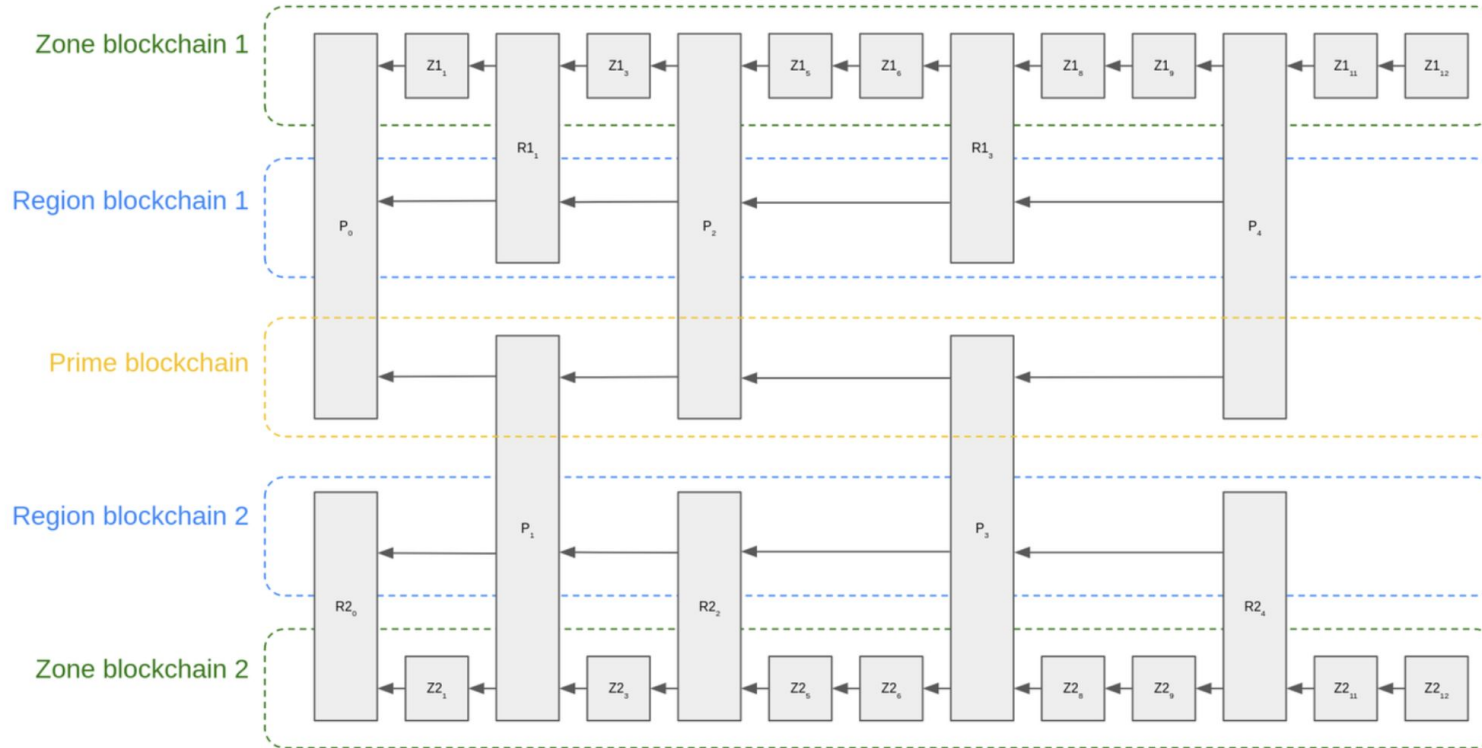
Hierarchical Sequences



Hierarchical Blockchains

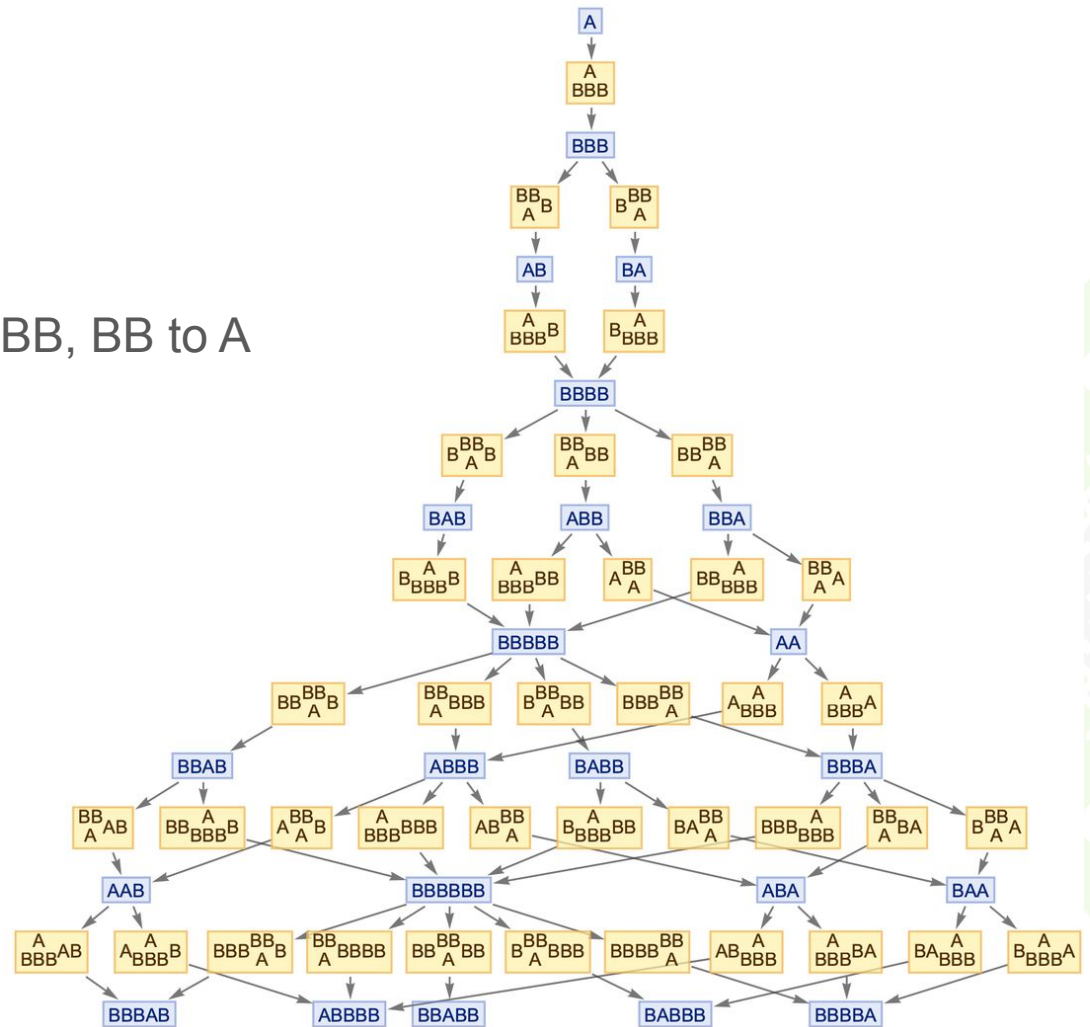


Quai Network



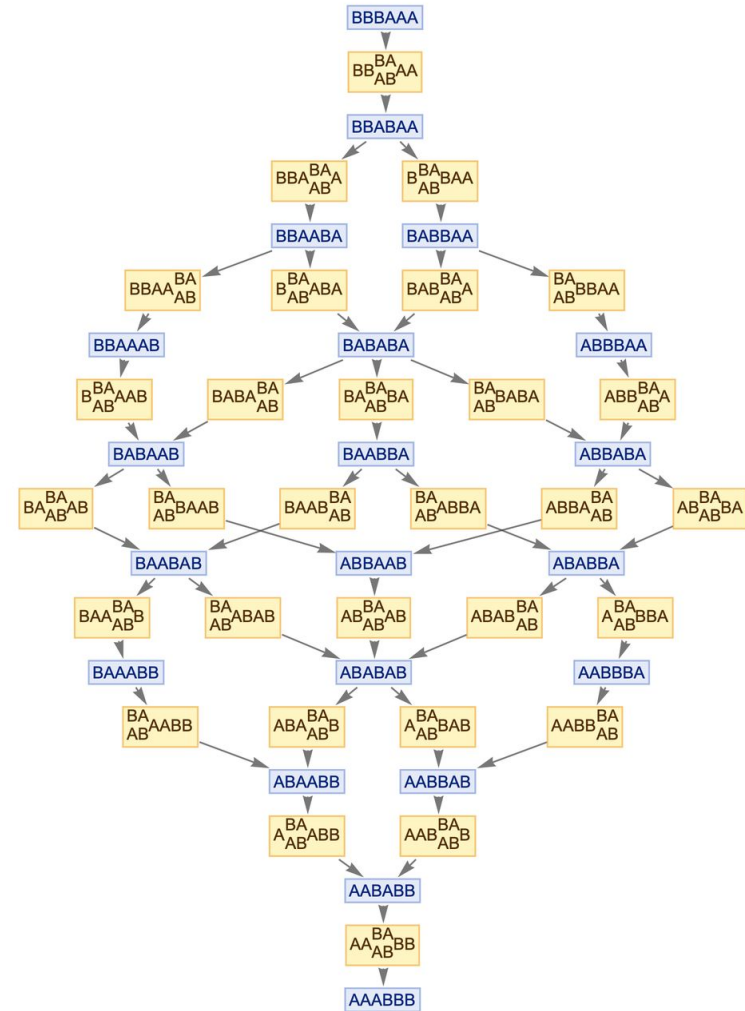
Physics

- Space of all possible rules
- Take one simple rule: A to BBB, BB to A
- Many transformations
- Multiway Hypergraph



Distributed Systems

- Branching & Merging
- Directed Acyclic Graph
- Resolution Events
- Consensus



Ontology, Epistemology, Computation

From Ontology to Computation A Structural Framework

Abstract

This paper provides a straightforward roadmap connecting physical reality to computational work. It argues that ontology, mathematics, epistemology, and computation form a sequential dependency. We establish how abstract reasoning and probabilistic data extraction are bounded by thermodynamic laws, framing computation as a physical process that requires energy to organize information.